

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fifth Semester B.E. (Biotechnology) (Credit System) Degree Examinations

January 2023

20BT502 / 17BT502 – ENZYME TECHNOLOGY

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **each Unit**.

Unit – I

- | | Marks | BT* | CO* | PO* |
|--|-------|-----|-----|-----|
| 1. a) Distinguish between chemical and enzymatic reactions. | 10 | L*4 | 1 | 1 |
| b) Define the function of enzymes. Explain about the nomenclature and classification of enzymes. | 10 | L1 | 1 | 1 |
| 2. a) Appraise the different hypothesis explaining enzyme-substrate interaction. | 10 | L5 | 1 | 1 |
| b) Discuss in detail about allosteric enzymes and coenzymes. | 10 | L2 | 1 | 1 |

Unit – II

- | | | | | |
|--|----|----|---|---|
| 3. a) Classify and explain the different types of catalysis with examples. | 10 | L2 | 2 | 1 |
| b) What are the factors affecting the kinetics of enzyme and explain in detail? | 10 | L1 | 2 | 1 |
| 4. a) Explain the below equation by using mathematical model assuming no product was present in the initial stage. | | | | |



- | | | | | |
|---|----|----|---|---|
| b) Explain the different types of inhibition with the help of examples. | 10 | L2 | 2 | 2 |
| | 10 | L2 | 2 | 1 |

Unit – III

- | | | | | |
|--|----|----|---|---|
| 5. a) Elaborate the various downstream processing steps involved for the separation of intracellular enzyme from microorganisms. | 10 | L4 | 3 | 2 |
| b) Discuss about the methods used for the isolation of pectinase from microorganisms. | 10 | L2 | 3 | 2 |
| 6. a) Explain about two different analytical techniques used for the determination of molecular weight of enzymes with suitable diagram. | 10 | L2 | 3 | 3 |
| b) What do you understand by the term High-throughput screening? | 10 | L2 | 3 | 3 |

Unit – IV

- | | | | | |
|---|----|----|---|---|
| 7. a) Comment about the industrial applications of immobilized enzymes. | 10 | L3 | 4 | 4 |
| b) Compare the various techniques used for immobilization of enzymes. | 10 | L4 | 4 | 3 |
| 8. a) Differentiate between fluidized bed reactor and CSTR. | 10 | L4 | 4 | 4 |
| b) How whole cells can be used for various industrial applications? | 10 | L1 | 4 | 4 |

Unit – V

- | | | | | |
|--|----|----|---|---|
| 9. a) How microorganisms can be used for the production of various chemicals? | 10 | L1 | 5 | 6 |
| b) Explain about the application of protease for various industrial processes. | 10 | L2 | 5 | 2 |
| 10. a) How enzyme activity determination can be used for detection of inborn errors of metabolism? | 10 | L5 | 5 | 4 |
| b) Write about identification and differentiation of isoenzymes. | 10 | L1 | 5 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
